DUKE KUNSHAN MATH UNION

Integration Bee 2026

Competition Problem Set

Difficulty-Rebalanced Edition

77 problems • Five competitive rounds • 20 backups

DKU MATH UNION

Stage	Problems	Time limit
Qualifier	10	20 minutes total
Round of 16	24	120 s / problem
Quarterfinals	12	150 s / problem
Semifinals	6	180 s / problem
Final	5	240 s / problem
Backups	20	Matched to stage

Contest questions only. No answer key or worked solutions included.

Source problem IDs and difficulty scores are recorded separately in the allocation CSV.

ROUND 01

Qualifier

10 problems • 20 minutes • Top 16 advance

PROBLEM 1

Evaluate $\displaystyle\int_0^1 \frac{dx}{1+x^2}$.

PROBLEM 2

Evaluate $\displaystyle\int_0^{\pi/2} \cos^2 x\, dx$.

PROBLEM 3

Evaluate $\displaystyle\int_0^{\pi} \sin^2 x\, dx$.

PROBLEM 4

Evaluate $\displaystyle\int_0^{\infty} xe^{-x^2}\, dx$.

PROBLEM 5

Evaluate $\displaystyle\int_0^{\pi/2} \frac{\cos x}{1+\sin^2 x}\, dx$.

PROBLEM 6

Evaluate $\displaystyle\int_0^1 \sqrt{1-x^2}\, dx$.

PROBLEM 7

Evaluate $\displaystyle\int_0^1 \frac{x}{1+x^4}\, dx$.

PROBLEM 8

Evaluate $\displaystyle\int_0^{\pi/2} \sin^2 x \cos^3 x\, dx$.

DKU MATH UNION

PROBLEM 9

Evaluate $\displaystyle\int_0^{\pi/4} \sec^4 x \, dx$.

PROBLEM 10

Evaluate $\displaystyle\int_0^1 x^3\sqrt{1-x^2}\, dx$.

DKU MATH UNION

ROUND 02

Round of 16

8 matches • Best of three • 120 seconds per problem

Game 1

One problem per match

PROBLEM 1

Evaluate $\int_0^1 xe^{-x}\,dx$.

PROBLEM 2

Evaluate $\int_0^{\pi/2} x\sin x\,dx$.

PROBLEM 3

Evaluate $\iint_D (x+y)\,dA$, where D is the triangle with vertices $(0,0)$, $(1,0)$, and $(0,1)$.

PROBLEM 4

Evaluate $\int_0^{\pi/2} \sin^2 x\cos^2 x\,dx$.

(Provided by Tianrui Gu)

PROBLEM 5

Evaluate $\int_0^1 \ln(1+x)\,dx$.

PROBLEM 6

Evaluate $\int_0^{\infty} xe^{-x}\,dx$.

PROBLEM 7

Evaluate $\int_1^{\infty} \frac{\ln x}{x^2}\,dx$ and determine whether it converges.

(Provided by Tianrui Gu)

DKU MATH UNION

PROBLEM 8

Evaluate $\displaystyle\int_0^1 \sqrt{x - x^2}\, dx$.

(Provided by Jiachen Rao)

Game 2

One problem per match

PROBLEM 9

Evaluate $\displaystyle\int_0^1 (\ln x)^2\, dx$.

PROBLEM 10

Evaluate $\displaystyle\int_0^{\pi/2} x \cos x\, dx$.

(Provided by Tianrui Gu)

PROBLEM 11

Evaluate $\displaystyle\iint_D (x^2 + y^2)\, dA$, where D is the unit disk $x^2 + y^2 \le 1$.

(Provided by Tianrui Gu)

PROBLEM 12

Evaluate $\displaystyle\int_1^e \frac{\ln x}{x^2}\, dx$.

PROBLEM 13

Evaluate $\displaystyle\oint_C (-y\, dx + x\, dy + z\, dz)$, where C is the unit circle in the xy-plane, oriented counterclockwise.

(Provided by Tianrui Gu)

PROBLEM 14

Evaluate $\displaystyle\int_0^1 \arctan x\, dx$.

PROBLEM 15

Evaluate $\displaystyle\int_0^1 x^2 e^{-x}\, dx$.

DKU MATH UNION

PROBLEM 16

Evaluate $\iint_D \sqrt{x^2+y^2}\,dA$, where D is the unit disk $x^2+y^2 \le 1$.

(Provided by Tianrui Gu)

Game 3

One problem per match

PROBLEM 17

Evaluate $\displaystyle\int_0^\infty \frac{dx}{(1+x^2)^2}$.

PROBLEM 18

Evaluate $\displaystyle\int_0^\infty \frac{\ln x}{1+x^2}\,dx$.

PROBLEM 19

Evaluate $\iint_D e^{x^2+y^2}\,dA$, where $D = \{(x,y) : x^2+y^2 \le 1,\ x \ge 0,\ y \ge 0\}$.

PROBLEM 20

Evaluate $\displaystyle\int_0^\infty e^{-x}\sin x\,dx$.

PROBLEM 21

Evaluate $\displaystyle\int e^x \cos x\,dx$.

PROBLEM 22

For $f(x) = x^2$ on $[-\pi,\pi]$, evaluate the cosine Fourier coefficient $a_n = \dfrac{1}{\pi}\displaystyle\int_{-\pi}^{\pi} x^2\cos(nx)\,dx$ for $n \ge 1$.

(Provided by Tianrui Gu)

PROBLEM 23

Find an antiderivative: $\displaystyle\int \frac{2x+1}{1+e^{x^2+x}}\,dx$.

(Provided by Jiachen Rao)

DKU MATH UNION

PROBLEM 24

Evaluate $\displaystyle\int_0^{\pi/2} \frac{\sin^4 x}{\sin^4 x + \cos^4 x}\, dx.$

(Provided by Jiachen Rao)

DKU MATH UNION

ROUND 03

Quarterfinals

4 matches • Best of three • 150 seconds per problem

Game 1

One problem per match

PROBLEM 1

Evaluate $\displaystyle\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x}\, dx$.

PROBLEM 2

Let E be the upper half of the unit ball. Evaluate $\displaystyle\iiint_E z\, dV$.

(Provided by Tianrui Gu)

PROBLEM 3

Evaluate $\displaystyle\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x}\, dx$.

(Provided by Tianrui Gu)

PROBLEM 4

Given that $\displaystyle\sum_{n\geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$, evaluate $\displaystyle\int_0^1 \frac{\ln(1-x)}{x}\, dx$.

(Provided by Tianrui Gu)

DKU MATH UNION

Game 2

One problem per match

PROBLEM 5

Given that $\sum_{n\geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$, evaluate $\int_0^1 \frac{\ln(1+x)}{x}\,dx$.

(Provided by Tianrui Gu)

PROBLEM 6

Evaluate the improper integral $\int_0^1 \left\lfloor \log_2 \frac{1}{x} \right\rfloor dx$.

(Provided by Jiachen Rao)

PROBLEM 7

Given that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$, evaluate $\int_0^1 \ln x \ln(1-x)\,dx$.

(Provided by Tianrui Gu)

PROBLEM 8

Evaluate $\int_0^1 \frac{\ln(1+x)}{1+x^2}\,dx$.

(Provided by Jiachen Rao)

Game 3

One problem per match

PROBLEM 9

Using a semicircular contour in the upper half-plane, evaluate $\int_{-\infty}^{\infty} \frac{x^2}{(x^2+1)(x^2+4)}\,dx$, justifying why the integral over the arc tends to 0.

(Provided by Luyao Xu)

PROBLEM 10

Evaluate $\int_0^{2\pi} \frac{\cos(3x)}{2+\cos x}\,dx$.

(Provided by Jiachen Rao)

PROBLEM 11

Evaluate $\int_0^1 \frac{\ln x}{1-x^2}\,dx$.

(Provided by Jiachen Rao)

DKU MATH UNION

PROBLEM 12

Evaluate $\displaystyle\int_0^\infty \frac{\sin x - x\cos x}{x^3}\,dx$.

(Provided by Tianrui Gu)

DKU MATH UNION

ROUND 04

Semifinals

2 matches • Best of three • 180 seconds per problem

Game 1

One problem per match

PROBLEM 1

Evaluate $\displaystyle\int_0^1 \frac{x^{2026}-1}{\ln x}\,dx$.

(Provided by Jiachen Rao)

PROBLEM 2

Evaluate $\displaystyle\int_0^\infty \frac{(\arctan x)^2}{x^2}\,dx$.

(Provided by Tianrui Gu)

Game 2

One problem per match

PROBLEM 3

For $x>0$, let $I(x)=\displaystyle\int_0^1 \frac{\ln(1+t^x)}{1+t}\,dt$. Express $I(x)$ as a series.

Hint. Expand $\ln(1+t^x)$ and $1/(1+t)$ as Taylor series, then integrate term by term.

(Provided by Haohong Yu)

PROBLEM 4

Using the convention $\widehat{f}(\omega)=\displaystyle\int_{-\infty}^{\infty} f(x)e^{-i\omega x}\,dx$, evaluate the Fourier transform of $f(x)=e^{-x^2}$:

$$\int_{-\infty}^{\infty} e^{-x^2}e^{-i\omega x}\,dx.$$

(Provided by Tianrui Gu)

Game 3

One problem per match

DKU MATH UNION

PROBLEM 5

Using contour integration, evaluate $\displaystyle\int_0^{2\pi} \frac{d\theta}{(5+3\cos\theta)^2}$.

(Provided by Luyao Xu)

PROBLEM 6

Evaluate $\displaystyle\int_0^{\infty} \left(\frac{\sin x}{x}\right)^3 dx$.

(Provided by Tianrui Gu)

DKU MATH UNION

ROUND 05

Final

1 match • Best of five • 240 seconds per problem

Game 1

Championship match

PROBLEM 1

Let $V \subset \mathbb{R}^3$ be a bounded region whose boundary $S = \partial V$ is a piecewise smooth closed surface with outward unit normal n, and let $\mathrm{r} = (x, y, z)$, $r = |\mathrm{r}|$. Prove that

$$\iint_S \frac{\mathrm{r} \cdot \mathrm{n}}{r^3}\, dS = \begin{cases} 4\pi, & \text{if the origin lies inside } V, \\ 0, & \text{if the origin lies outside } V. \end{cases}$$

(Provided by Tianrui Gu)

Game 2

Championship match

PROBLEM 2

Let $f \in C^1[0,1]$ satisfy $f(0) = 0$ and $f'(x) > 0$ for $0 < x \le 1$. Prove that

$$\int_0^1 \frac{x}{f(x)}\, dx \le 2 \int_0^1 \frac{dx}{f'(x)}$$

and that the constant 2 is best possible.

(Provided by Haohong Yu)

Game 3

Championship match

PROBLEM 3

Let $0 < \alpha < 1$. Using a keyhole contour and the residue theorem, evaluate $I(\alpha) = \displaystyle\int_0^\infty \frac{x^{\alpha-1}}{1+x}\, dx$.

(Provided by Luyao Xu)

Game 4

Championship match

DKU MATH UNION

PROBLEM 4

For $0 \le x \le 1$, define the infinite nested radical by $R(x) = \lim_{N\to\infty} r_1^{(N)}$, where $r_{N+1}^{(N)} = 0$ and

$$r_n^{(N)} = \sqrt{1 + (x+n) r_{n+1}^{(N)}} \quad (n = N, \dots, 1).$$

Evaluate $\displaystyle\int_0^1 \frac{dx}{1 + R(x)^2}$.

(Provided by Jiachen Rao)

Game 5

Championship match

PROBLEM 5

For $n \ge 1$, let $E_k = E_k(x_1, \dots, x_n)$ denote the kth elementary symmetric polynomial, and set $A_n = E_1 - E_3 + E_5 - \cdots$ and $B_n = E_2 - E_4 + E_6 - \cdots$, where terms with indices greater than n are omitted. Evaluate

$$I_n = \int_{\mathbb{R}^n} \frac{\log(A_n^2 + B_n^2)}{\prod_{j=1}^n (1 + x_j^2)}\, dx_1 \cdots dx_n.$$

(Provided by Haohong Yu)

DKU MATH UNION

RESERVE POOL

Backup Problems

20 reserved problems • Select a replacement of comparable difficulty

Qualifier backup pool

3 problems

PROBLEM 1

Evaluate $\displaystyle\int_0^1 x^2\,dx$.

PROBLEM 2

Evaluate $\displaystyle\int_0^{\pi/2} \sin x\,dx$.

PROBLEM 3

Evaluate $\displaystyle\int_0^{\pi/3} \sec^2 x\,dx$.

Round of 16 backup pool

7 problems

PROBLEM 4

Evaluate $\displaystyle\int_0^{\infty} \frac{dx}{1+x^2}$.

PROBLEM 5

Evaluate $\displaystyle\int \frac{dx}{x\ln x}$, for $x > 1$.

PROBLEM 6

Evaluate $\displaystyle\int_0^{\pi} |\cos x|\,dx$.

PROBLEM 7

Evaluate $\displaystyle\int_0^1 x^2 \ln x\,dx$.

DKU MATH UNION

(Provided by Tianrui Gu)

PROBLEM 8

Let S be the sphere of radius a centered at the origin, oriented outward, and let $\mathrm{F}(x, y, z) = (x, y, z)$. Evaluate $\iint_S \mathrm{F} \cdot \mathrm{n}\, dS$.

(Provided by Tianrui Gu)

PROBLEM 9

Let $\mathrm{F}(x, y, z) = (x, y, z)$ and let S be the sphere of radius 2 centered at the origin, oriented outward. Evaluate $\iint_S \mathrm{F} \cdot \mathrm{n}\, dS$.

(Provided by Tianrui Gu)

PROBLEM 10

Evaluate $\oint_C (x\,dy - y\,dx)$, where C is the unit circle oriented counterclockwise.

(Provided by Tianrui Gu)

Quarterfinals backup pool

5 problems

PROBLEM 11

Given that $\displaystyle\int_0^\infty \frac{\sin u}{u}\,du = \frac{\pi}{2}$, evaluate $\displaystyle\int_0^\infty \frac{\sin^2 x}{x^2}\,dx$.

(Provided by Tianrui Gu)

PROBLEM 12

Evaluate $\displaystyle\int_0^1 \int_x^1 e^{y^2}\,dy\,dx$.

(Provided by Tianrui Gu)

PROBLEM 13

Evaluate the Laplace transform $\mathcal{L}\{\sin t\}(s) = \displaystyle\int_0^\infty e^{-st} \sin t\,dt$ for $s > 0$.

PROBLEM 14

Let S be the upper hemisphere of radius a centered at the origin, $z \geq 0$. Evaluate $\iint_S z\,dS$.

(Provided by Tianrui Gu)

DKU MATH UNION

PROBLEM 15

Given that $\displaystyle\sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{\pi^2}{6}$, evaluate $\displaystyle\int_0^{\infty}\frac{x}{e^x-1}\,dx$.

(Provided by Tianrui Gu)

Semifinals backup pool

3 problems

PROBLEM 16

Evaluate $\displaystyle\int_0^{\infty}\frac{dx}{1+x^4}$.

(Provided by Jiachen Rao)

PROBLEM 17

Evaluate $\displaystyle\int_0^{\infty}\frac{\sin x}{x}\,dx$ using the Laplace transform.

(Provided by Tianrui Gu)

PROBLEM 18

For a real constant a with $a\neq\pm1$, evaluate $\displaystyle\int_0^{\pi}\ln(1-2a\cos x+a^2)\,dx$.

(Provided by Tianrui Gu)

Final backup pool

2 problems

PROBLEM 19

Evaluate $\displaystyle\int_0^{\pi/2}\ln^2(\sin x)\,dx$.

(Provided by Tianrui Gu)

PROBLEM 20

Let $f\in C^1[0,1]$ with $f(0)=f(1)=0$. Prove that

$$\int_0^1 f(x)^2\,dx\le\frac{1}{\pi^2}\int_0^1 f'(x)^2\,dx,$$

and determine every f for which equality holds.

(Provided by Tianrui Gu)

DKU MATH UNION